



Diversity-Enhanced and Classification-Aware Prompt Learning for Few-Shot Learning via Stable Diffusion

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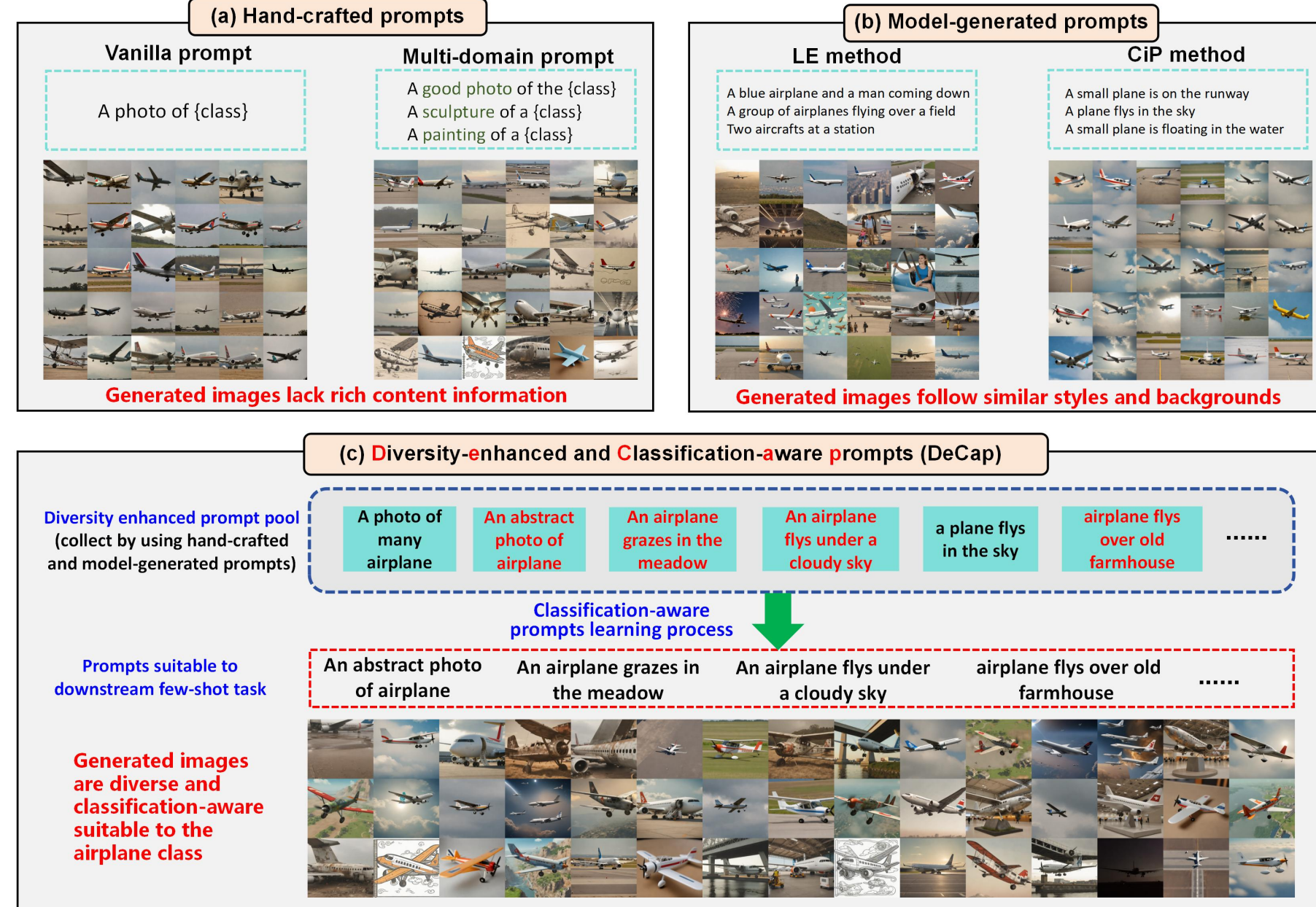
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Introduction

Motivation

Synthetic datasets generated by large scale generative models have the potential to alleviate data scarcity in few-shot classification problem. However, how to design task-specific prompts to produce suitable synthetic datasets remains a key research focus. Existing methods are often limited to design hand-crafted or model-generated prompts, which can lead to issues such as insufficient diversity and weak relevance to the classification task.

Prompt diversity and task relevance



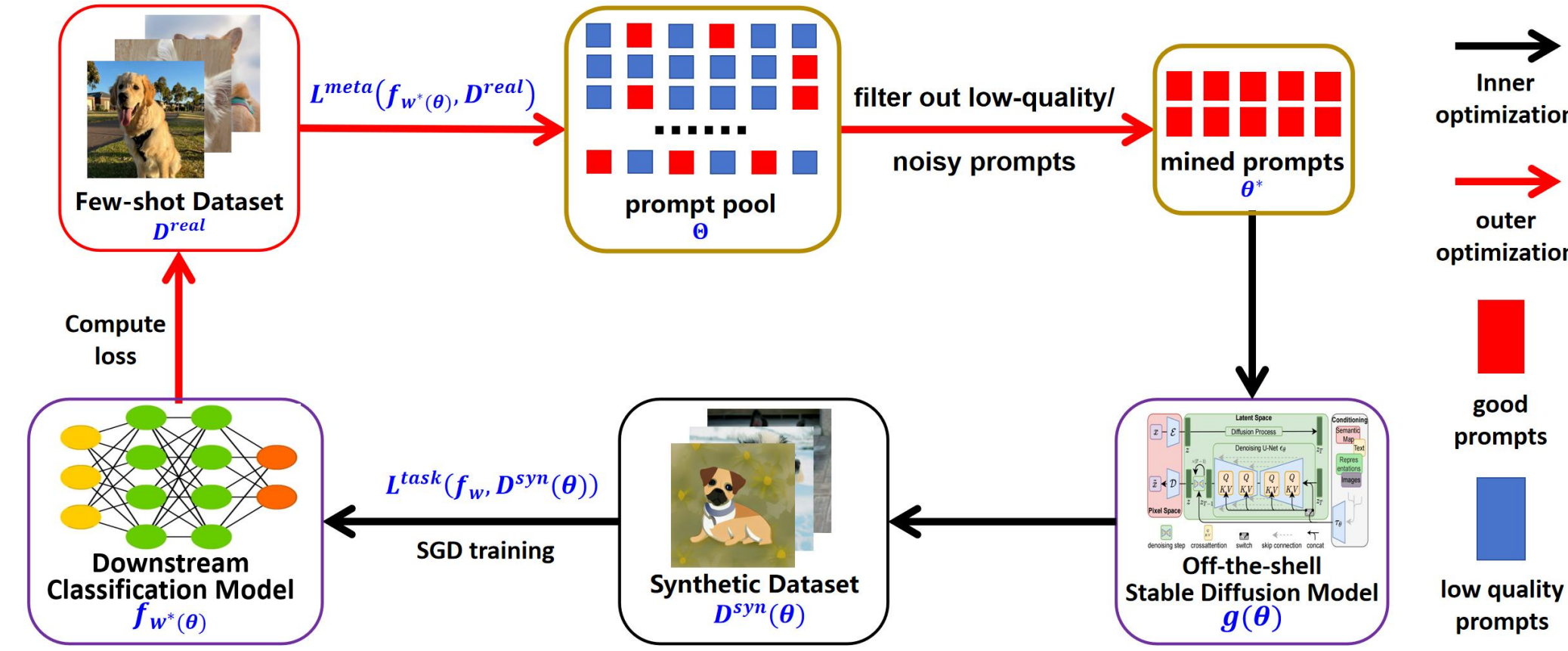
Hand-crafted prompts emphasize domain variation; model-generated prompts add content. DeCap combines both and learns task-relevant subsets.

Main contributions

- Learn discrete prompts with a classification-aware meta-objective.
- Adaptively filter noisy labels, caption errors and low-quality prompts.
- Balance shared domain templates with class-specific content descriptions.
- Improve existing zero/few-shot pipelines across the evaluated architectures.

Method

Two coupled learning loops



Diversity-enhanced prompt pool

Shared hand-crafted templates provide domain variation. Class-specific LE and CiP descriptions provide richer content.

$$\theta_i = [\theta_i^h, \theta_i^m]$$

Classification-aware selection

$$\theta^* = \arg \min_{\theta} \mathcal{L}^{\text{meta}}(f_{w^*}(\theta)(X^{\text{real}}), Y^{\text{real}})$$

$$w^* = \arg \min_w \mathcal{L}^{\text{task}}(f_w(X_g^{\text{syn}}(\theta)), Y^{\text{syn}})$$

- Inner loop: train the classifier on synthetic images generated by the selected prompts.
- Outer loop: evaluate on real few-shot images to guide the next prompt selection.
- Real few-shot supervision evaluates the classifier and guides selection from a diverse prompt pool.

Discrete search and implementation

1. Encode candidate prompt sets as integer indices.
2. Generate synthetic data, train the classifier, and evaluate the meta-loss on real few-shot images.
3. Update candidates by genetic search.

Algorithm 1 Learning Algorithm of the DeCap Method

```

1: GA.initial(max-iter, popsize)
2: for iter = 1, 2, ..., max-iter do
3:   fit = [], Pop = []
4:   for m = 1, 2, ..., popsize do
5:     pop^(m) = GA.sample()
6:     theta^(m), Y^syn = get_prompt(pop^(m), pool)
7:     X^syn(theta^(m)) = g(theta^(m))
8:     w*(theta^(m)) = arg min_w L^task(f_w(X^syn(theta^(m))), Y^syn)
9:     fit_pop^(m) = L^meta(f_w*(theta^(m)), D^real)
10:    fit.append(fit_pop^(m)), Pop.append(pop^(m))
11:  end for
12:  GA.update(fit, Pop)
13: end for

```

Algorithm 2 Get_prompt Algorithm

Input: indexes pop , prompt pool $pool$;
Output: prompt set: prompts, labels: Y^{syn} .

```

1: pop.reshape[N, M]
2: prompts = [], Y^syn = []
3: for i = 1, 2, ..., N do
4:   class = class_names[i]
5:   prompts.append(pool[class][pop[i]])
6:   Y^syn.append(i.repeat[M])
7: end for
8: return prompts, Y^syn

```

Results

Classification performance

12 datasets; 10 real images per class guide prompt selection. Downstream training uses synthetic-only or synthetic + real data.

	STL-10	CIFAR10	Imagenette	Pets	Caltech-101	Imagenet100	EuroSAT	Aircraft	Country211	DTD	UCF101	Imagenet
<i>without real zero-shot</i>	94.26	70.25	97.22	81.85	83.89	70.14	23.11	17.07	13.44	42.39	60.80	58.18
vanilla prompt	95.33	72.37	97.69	82.29	84.74	70.62	31.31	17.04	13.72	49.20	60.83	58.45
multi-domain	94.97	70.66	97.89	83.07	87.56	70.50	30.11	17.85	13.90	49.63	60.83	58.64
LE	94.61	70.33	97.45	83.24	84.03	70.73	29.35	17.73	14.14	52.50	61.06	58.86
CiP	94.92	70.24	97.65	84.04	88.12	70.76	39.91	18.00	14.98	60.69	63.63	59.75
DeCap (ours)	95.91	76.98	97.95	85.36	88.67	71.08	41.94	19.74	15.44	62.39	64.45	60.29
<i>with real real-only</i>	94.28	70.33	97.22	81.96	84.51	70.33	24.15	19.41	13.80	42.55	61.09	58.51
vanilla prompt	95.55	76.20	98.00	83.84	89.85	70.87	47.83	18.21	13.77	51.97	60.85	59.12
multi-domain	95.02	74.54	97.92	84.56	90.31	70.62	43.30	18.99	13.90	51.76	60.90	59.17
LE	94.72	71.66	97.49	84.00	84.34	70.46	42.06	20.22	14.37	54.04	61.54	59.85
CiP	95.05	70.51	97.75	85.16	89.86	70.86	49.17	20.31	15.49	62.66	64.66	62.51
DeCap (ours)	95.93	77.19	98.03	85.78	89.87	71.11	50.22	20.64	15.68	63.99	65.21	62.82

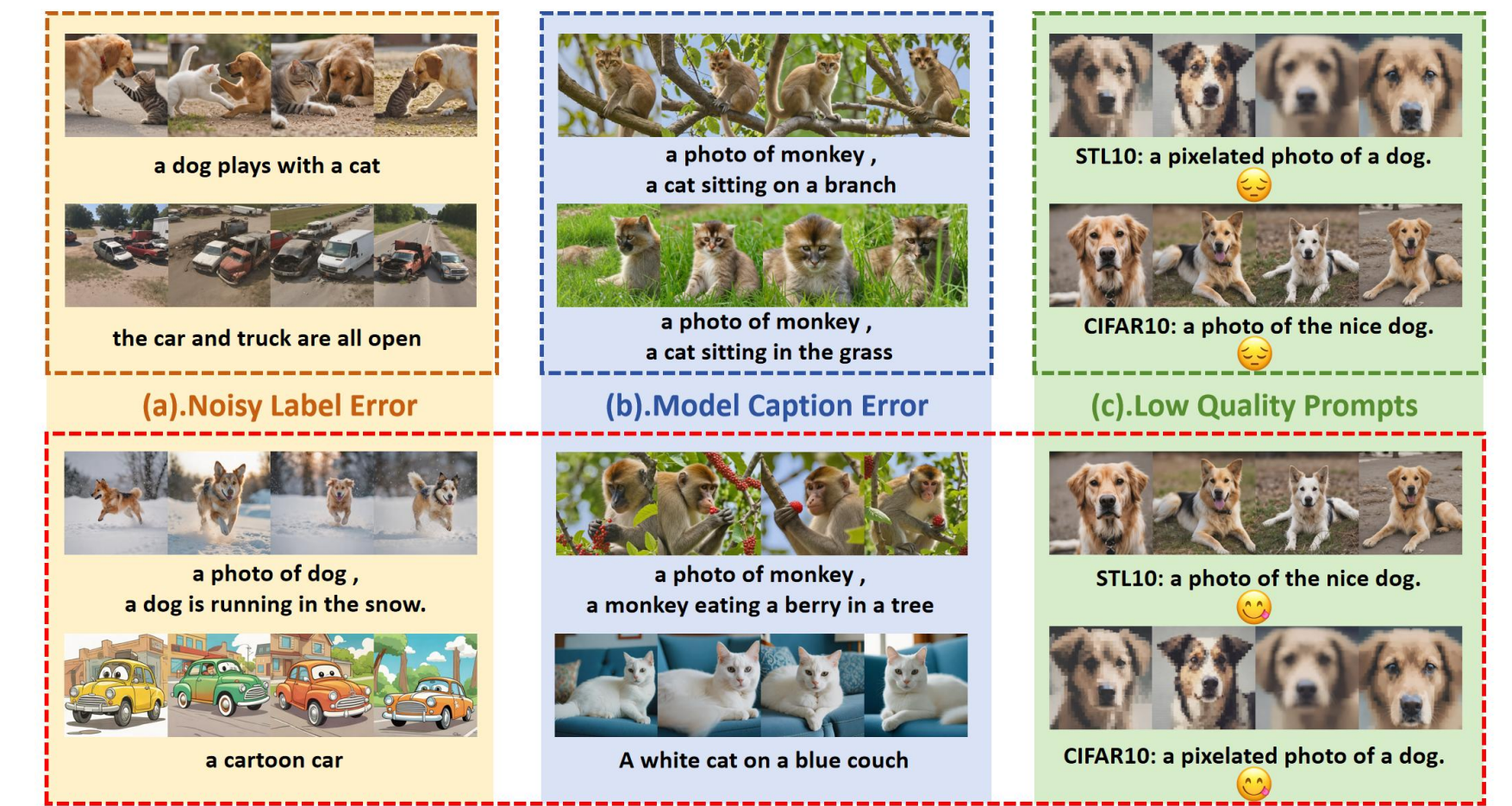
Best synthetic-only accuracy on all 12 datasets (Table 1). Metric: top-1 accuracy (%).

Across algorithms and architectures

	STL-10	CIFAR10	Imagenette	Pets	Caltech-101	Imagenet100	EuroSAT	Aircraft	Country211	DTD	UCF101	Imagenet
Fakelt	52.26	38.45	69.60	29.74	66.20	32.75	48.40	37.70	3.61	31.25	33.30	15.46
With DeCap	60.39	48.80	75.40	55.22	70.51	39.21	51.20	40.60	4.22	41.67	41.07	21.59
SuS-X	95.24	72.77	98.24	79.64	84.57	69.96	33.89	18.30	12.97	46.65	57.18	57.84
With DeCap	95.43	75.89	98.39	80.40	84.89	70.3	37.37	19.83	13.63	56.49	61.8	59.54
CaFo	95.33	85.34	97.66	86.62	94.09	74.64	83.5	26.07	16.20	65.74	72.69	63.41
With DeCap	95.90	86.00	98.06	88.66	94.28	76.28	85.59	32.10	16.88	68.03	75.02	64.29

Replacing data generation improves Fakelt, SuS-X and CaFo in the evaluated settings. CaFo uses 16 shots (Table 2).

Prompt quality and task adaptation



Class-specific prompt distributions

